Arista Access Node QoS/Congestion Event Troubleshooting and Resolution SOP

Node: MAR-ACC-45 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_1079

Description

The Arista access node, situated at the Marathahalli Core site, designated as MAR-ACC-45, is prone to QoS/Congestion events such as microbursts, interface tail drops, and QoS queue starvation. This is predominantly due to the intrinsic mechanics of packet switching and queue management within the network equipment. The Arista EOS (Extensible Operating System) manages traffic flows through a combination of ingress and egress buffering, alongside a sophisticated QoS framework that prioritizes and shapes traffic. However, under certain conditions, such as sudden spikes in traffic volume or misconfigured QoS policies, the buffers can overflow, leading to packet drops and congestion. The role of the access node, being the first point of contact for user traffic, exacerbates this issue as it aggregates traffic from multiple sources. Thus, understanding the underlying protocol and hardware mechanics is crucial for troubleshooting and resolving these events efficiently.

Pre-requisite Checks

- Confirm redundant link status to ensure no single point of failure
- Dispatch remote hands to Marathahalli Core for potential hardware intervention
- Verify power supply and environmental conditions at the site

Potential Root Causes

Cause	Probability
Misconfigured QoS Policies	High
Insufficient Buffer Allocation	Medium
Hardware Failure (e.g., Faulty Transceiver)	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos interface ethernet 1, Expected Output: QoS configuration and statistics for Ethernet 1, Description: This command isolates the QoS configuration applied to the specified interface, providing insights into potential misconfigurations that could lead to congestion events.

Command: show interface ethernet 1 counters, Expected Output: Interface counters including errors, drops, and packets, Description: By examining the interface counters, one can identify if there are any anomalies in packet processing that could indicate congestion or buffer overflow issues.
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```
Command: show buffers, Expected Output: Buffer allocation and utilization statistics, Description: Analyzing buffer utilization helps in determining if the allocated buffers are sufficient for the current traffic profile, or if adjustments are needed to prevent overflow.
```

Resolution Steps

Step 1: Step: Adjust QoS Policies, Command: interface ethernet 1 qos trust dscp, Description: Adjusting the QoS trust settings to ensure proper prioritization of traffic based on DSCP markings can help mitigate congestion by ensuring critical traffic is not dropped.

Step 2: Step: Allocate Additional Buffers, Command: buffer-size ethernet 1 1024, Description: Increasing the buffer size on congested interfaces can provide a temporary relief by allowing the network to absorb sudden spikes in traffic, reducing the likelihood of drops.

Step 3: Step: Implement Traffic Shaping, Command: policy-map shape-traffic class class-default police cir 100m, Description: Implementing traffic shaping policies can help regulate the amount of traffic allowed into the network, preventing overwhelming of the access node and subsequent congestion events.

Post-Check Verification

```
Command: show qos interface ethernet 1, Expected Output: Updated QoS configuration and reduced drop rates, Description: Re-running the QoS interface command after adjustments to verify that the changes have taken effect and are mitigating the congestion issue.

Command: show interface ethernet 1 counters, Expected Output: Reduced error and drop counters, Description: Monitoring interface counters post-implementation to ensure that the applied fixes have resulted in a decrease in errors and dropped packets.
```

Roll-back Procedures

- Step: Revert QoS Configuration, Command: no interface ethernet 1 qos trust dscp, Description: If adjustments to QoS policies do not yield the desired outcome or introduce new issues, reverting these changes is essential to restore the original state.

- Step: Revert Buffer Allocation, Command: no buffer-size ethernet 1 1024, Description: If increased buffer allocation does not alleviate congestion or introduces performance issues, reverting to the original buffer size is necessary.